

Assignment 2 - EE4725 Quasi Optical Systems

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Contents

1	System Parameters	2
2	Q1: Feed Far Field (1 point)	2
3	Q2: Aperture Current Distribution, $f/D = 2$ (2 points)	3
4	Q3: Reflector Far Field, $f/D = 1$ (2 points)	5
5	Q4: Efficiencies vs D (3 points)	6
6	Q5: Directivity and Gain vs D (2 points)	7

1 System Parameters

The system under analysis is a front-fed parabolic reflector antenna with the following parameters:

- Operating frequency: $f = 500$ GHz, wavelength $\lambda = 0.6$ mm, wavenumber $k_0 = 2\pi/\lambda$
- Feed: circular aperture of diameter $D_f = 4\lambda = 2.4$ mm (radius $a = 2\lambda$), with a $\hat{y}$ -polarized uniform electric current distribution
- Parabolic reflector focal length: $f = 5$ m
- Free-space impedance: $\zeta = 120\pi \Omega$

2 Q1: Feed Far Field (1 point)

The normalized far field patterns of the circular aperture feed are computed in two principal planes:

- **E-plane** ($\phi = 90^\circ$): contains the $\hat{y}$ -axis (current polarization direction)
- **H-plane** ($\phi = 0^\circ$): perpendicular to the current direction

The far field shape is governed by the Airy diffraction pattern, arising from the 2D Fourier transform of the uniform circular aperture:

$$\tilde{J}(k_\perp) = \pi a^2 \cdot \frac{2 J_1(k_\perp a)}{k_\perp a} \quad (1)$$

where J_1 is the Bessel function of the first kind and $k_\perp = k_0 \sin \theta$. The first null occurs at:

$$\theta_{\text{null}} = \arcsin\left(\frac{1.22 \lambda}{D_f}\right) = \arcsin\left(\frac{1.22}{4}\right) = 17.76^\circ \quad (2)$$

For a $\hat{y}$ -directed current, the relevant SGF column is (G_{xy}, G_{yy}, G_{zy}) , and the far field is obtained via stationary phase evaluation.

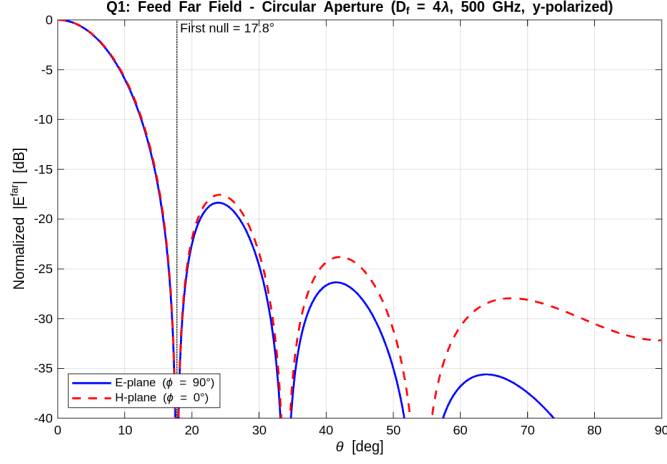


Figure 1: Normalized far-field pattern of the circular aperture feed ($D_f = 4\lambda$, 500 GHz, $\hat{y}$ -polarized) in the E-plane and H-plane.

Observations from Figure 1:

- The computed first null at $\theta = 17.8^\circ$ matches the theoretical value of 17.76° .
- The E-plane and H-plane patterns share the same main lobe width but differ in their sidelobe structure. In the H-plane, the SGF introduces an additional $\cos \theta$ factor on the dominant E_θ component, resulting in higher sidelobes compared to the E-plane. Conversely, the E-plane shows deeper nulls between sidelobes.
- This E/H asymmetry is a fundamental property of linearly polarized aperture antennas: the two planes experience different projections of the vector radiation pattern.

3 Q2: Aperture Current Distribution, $f/D = 2$ (2 points)

For $f/D = 2$, the reflector diameter is $D = 2.5$ m and the half-subtended angle is:

$$\theta_0 = 2 \arctan\left(\frac{D}{4f}\right) = 14.25^\circ \quad (3)$$

The aperture field $\vec{E}_a$ is obtained by evaluating the feed far field at angles (θ', ϕ') corresponding to each aperture point via the paraboloid mapping $\rho' = 2f \tan(\theta'/2)$, then applying the GO amplitude taper $(1 + \cos \theta')/(2f)$.

The equivalent aperture currents are computed on a 201×201 Cartesian grid using Love's equivalence principle:

$$\vec{M}_s = -\hat{z} \times \vec{E}_a, \quad \vec{J}_s = \hat{z} \times \vec{H}_a = -\frac{1}{\zeta} \vec{E}_a \quad (4)$$

giving $M_x = E_{a,y}$, $M_y = -E_{a,x}$, $J_x = -E_{a,x}/\zeta$, $J_y = -E_{a,y}/\zeta$.

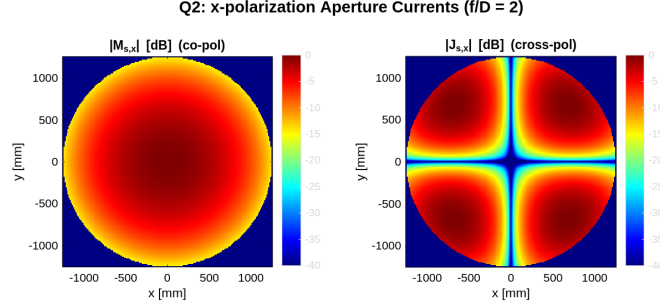


Figure 2: x -polarization aperture currents in dB: magnetic current $|M_{s,x}|$ (co-pol) and electric current $|J_{s,x}|$ (cross-pol) for $f/D = 2$.

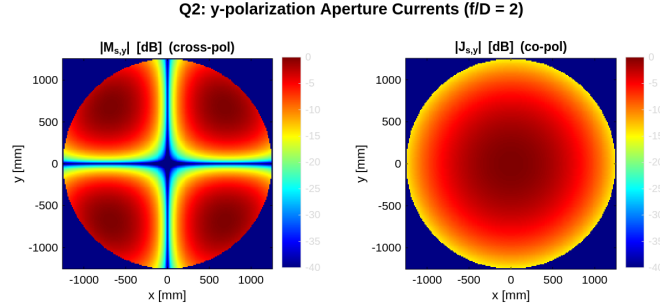


Figure 3: y -polarization aperture currents in dB: magnetic current $|M_{s,y}|$ (cross-pol) and electric current $|J_{s,y}|$ (co-pol) for $f/D = 2$.

Observations from Figures 2 and 3:

- **Co-pol components** ($M_x = E_{a,y}$ and $J_y = -E_{a,y}/\zeta$): show a smooth, nearly circularly symmetric amplitude distribution with a gradual 3–4 dB taper from center to edge. This taper arises from the combination of the feed pattern roll-off and the GO spreading factor $(1 + \cos \theta')/(2f)$.
- **Cross-pol components** ($M_y = -E_{a,x}$ and $J_x = -E_{a,x}/\zeta$): display a characteristic quadrupolar (four-lobed) pattern with deep nulls along the

$x = 0$ and $y = 0$ planes. The peak cross-pol level is roughly 15–20 dB below the co-pol peak. This pattern arises from the SGF coupling between the $\hat{y}$ -directed feed current and the x -component of the radiated field.

4 Q3: Reflector Far Field, $f/D = 1$ (2 points)

For $f/D = 1$, $D = 5$ m, and $\theta_0 = 2 \arctan(1/4) = 28.07^\circ$. The aperture currents are computed on a 301×301 grid, and the far field is obtained via numerical 2D Fourier transform using the Balanis spatial-domain formulation:

$$E_\theta \propto (-L_x \sin \phi + L_y \cos \phi) + \zeta (N_x \cos \phi + N_y \sin \phi) \cos \theta \quad (5)$$

$$E_\phi \propto (L_x \cos \phi + L_y \sin \phi) \cos \theta + \zeta (N_x \sin \phi - N_y \cos \phi) \quad (6)$$

where $\vec{N} = \iint \vec{J}_s e^{j(k_x x' + k_y y')} dx' dy'$ and $\vec{L} = \iint \vec{M}_s e^{j(k_x x' + k_y y')} dx' dy'$.

The beam is extremely narrow: the theoretical Airy null is at $\arcsin(1.22\lambda/D) = 0.0084^\circ$ for $D = 5$ m at $\lambda = 0.6$ mm.

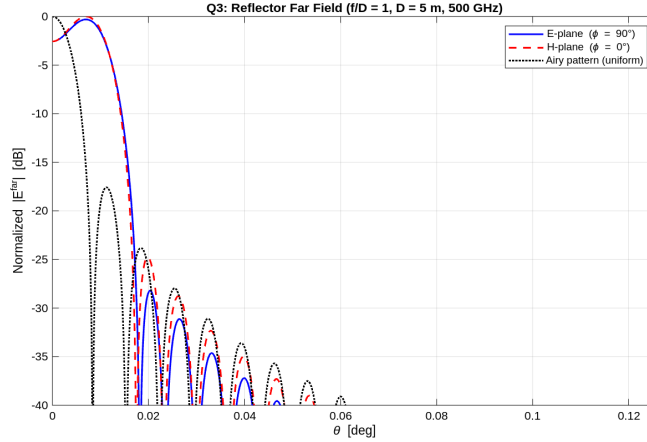


Figure 4: Normalized far-field pattern of the parabolic reflector ($f/D = 1$, $D = 5$ m) in the E-plane and H-plane, compared to the Airy pattern of a uniformly illuminated aperture.

Observations from Figure 4:

1. **Wider main beam:** The reflector pattern has its first null at approximately 0.01° , compared to 0.0084° for the Airy pattern. This widening is due to the amplitude taper across the aperture — the edges are less illuminated, reducing the effective aperture size.
2. **Lower sidelobes:** The first sidelobe of the reflector pattern is approximately -25 dB, significantly below the Airy pattern's first sidelobe at

−17.6 dB. The amplitude taper acts as a natural window function that suppresses sidelobes.

3. **E/H plane similarity:** The E-plane and H-plane patterns are very similar, with only minor differences in the sidelobe region. This is expected because the co-pol aperture distribution is nearly circularly symmetric.

5 Q4: Efficiencies vs D (3 points)

Three efficiencies are computed as a function of reflector diameter D for $0.6 < f/D < 6$. The spillover efficiency is:

$$\eta_s = \frac{\int_0^{\theta_0} \int_0^{2\pi} U_{\text{feed}} \sin \theta' d\theta' d\phi'}{\int_0^{\pi/2} \int_0^{2\pi} U_{\text{feed}} \sin \theta' d\theta' d\phi'} \quad (7)$$

The taper efficiency, using the Ludwig-3 co-pol definition $E_{\text{co}} = E_\theta \sin \phi + E_\phi \cos \phi$, is:

$$\eta_t = \frac{16}{\pi} \left(\frac{f}{D} \right)^2 \frac{\left| \int_0^{\theta_0} \int_0^{2\pi} |E_{\text{co}}| \tan\left(\frac{\theta'}{2}\right) d\theta' d\phi' \right|^2}{\int_0^{\theta_0} \int_0^{2\pi} G(\theta', \phi') \sin \theta' d\theta' d\phi'} \quad (8)$$

The aperture efficiency is $\eta_{\text{ap}} = \eta_s \cdot \eta_t$.

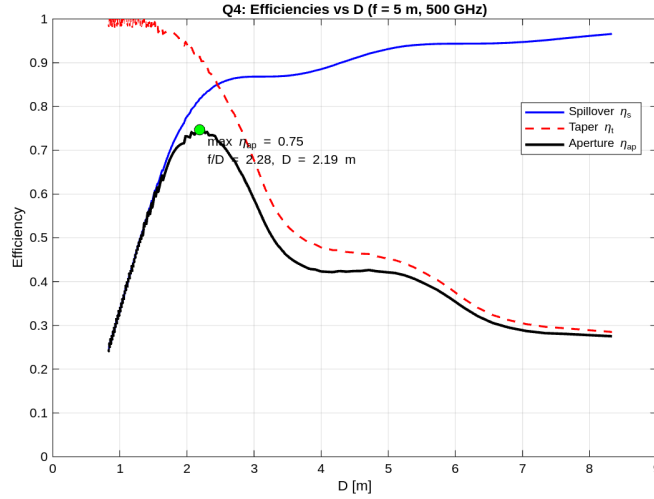


Figure 5: Spillover, taper, and aperture efficiency vs. reflector diameter D ($f = 5$ m, 500 GHz).

Observations from Figure 5:

- **Large D (small f/D , deep dish):** The reflector subtends a large angle (θ_0 up to $\sim 45^\circ$ at $f/D = 0.6$). Spillover efficiency η_s is high (~ 0.96) because the large dish captures most of the feed power. However, taper efficiency η_t is poor (~ 0.3) because the feed pattern has already passed through its first null at 17.8° and the illumination is highly non-uniform.
- **Small D (large f/D , shallow dish):** Taper efficiency approaches 1.0 because only the smooth central portion of the feed pattern illuminates the dish. However, spillover efficiency drops significantly (below 0.5 at $f/D > 4$) as most feed power misses the small reflector.
- **Optimal D :** The aperture efficiency peaks at $\eta_{ap} = 0.75$ at $f/D = 2.28$ ($D = 2.19$ m), where $\eta_s = 0.87$ and $\eta_t = 0.86$. This corresponds to an edge taper of roughly -10 to -12 dB, the well-known criterion for maximum aperture efficiency in reflector antenna design.

6 Q5: Directivity and Gain vs D (2 points)

The maximum directivity, directivity, and gain are:

$$D_{\max} = \left(\frac{\pi D}{\lambda} \right)^2, \quad D = \eta_t D_{\max}, \quad G = \eta_s \eta_t D_{\max} \quad (9)$$

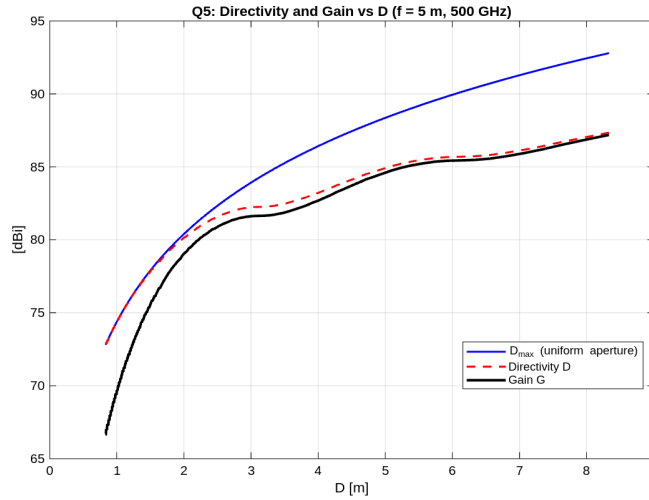


Figure 6: Maximum directivity $D_{\max}$, directivity D , and gain G vs. reflector diameter D in dBi ($f = 5$ m, 500 GHz).

Observations from Figure 6:

- $D_{\max}$ increases monotonically with D , following the $(\pi D/\lambda)^2$ scaling. At $D = 5$ m, $D_{\max} \approx 88.4$ dBi. This is purely geometric — larger aperture means more collecting area.
- **Directivity** D is reduced from $D_{\max}$ by the taper efficiency. At large D (small f/D), the taper loss is substantial (~ 5 – 6 dB) because the feed pattern illuminates the aperture very non-uniformly. At small D (large f/D), D approaches $D_{\max}$ as taper efficiency is near unity.
- **Gain** G is further reduced by spillover loss. At small D (large f/D), the gap between G and D is large (up to 5 + dB) because most feed power misses the small reflector. At large D (small f/D), G is close to D since the big dish captures nearly all feed power ($\eta_s \approx 1$).
- The relationship $G \leq D \leq D_{\max}$ always holds. The gap $D_{\max} - G$ in dB equals $-10 \log_{10}(\eta_{\text{ap}})$, which is minimized at the optimal $f/D = 2.28$ where $\eta_{\text{ap}} = 0.75$ (loss of only 1.25 dB).